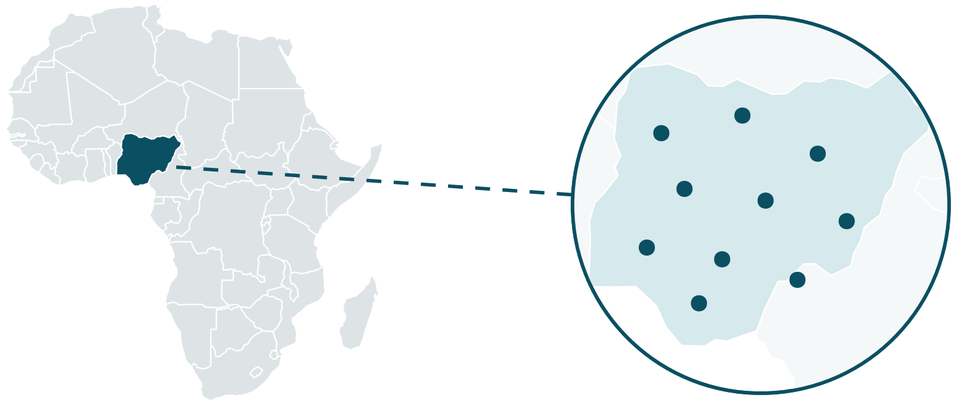
Genomic surveillance of *pfhrp2* and *pfhrp3* deletion and rapid diagnostic test (RDT) implications, Nigeria

A retrospective molecular surveillance study was conducted to determine the prevalence of *pfhrp2* and *pfhrp3* deletions in *P. falciparum* isolates collected from RDT false-negative and true-positive cases across Nigeria.



Total samples tested

66

31 RDT false-negative

35 RDT true-positive

Deletion detected

pfhrp3 deletion in 4 samples

(all also carried *pfhrp2* deletion)

None among RDT true positives

Approach

Nested PCR targeting *pfhrp2* and *pfhrp3* on 196 samples (31 RDT false-negative, 35 RDT true-positive, and other controls), followed by Sanger sequencing.

Key findings

pfhrp3 deletion was detected in 4 samples, all of which also had *pfhrp2* deletions. No deletions were observed among RDT true-positive cases.

Action supported

Findings support routine molecular monitoring to detect emerging deletions and guide the use of RDTs in Nigeria.

Relevance to this section

This case study demonstrates how genomic surveillance detects *pfhrp2*/*pfhrp3* deletions and informs diagnostic policy decisions.

Full record

Ethics, DBS handling, PCR and deletion-calling criteria, repeat-type analysis and metadata fields: Case Study 3, Section 13.

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